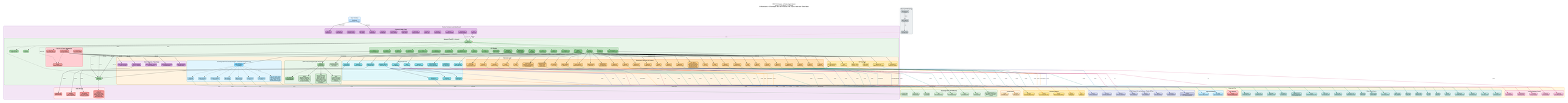

ABCT Architecture - A Better Crypto Tracker

Version 1.15.2 (Build 1771794638)

53 Blockchains | 42 Exchanges | 80+ DeFi Protocols | P&L Engine | Multi-User | Demo Mode

User Interface
Web Browser
localhost:8000 / Port 8080
All views (HTTP GET)
Authentication
Docker Container: abct-dashboard
Frontend (Static Files)
index.html
Dashboard UI
wallets.html
Wallet Manager
transactions.html
Transaction History
nft-wall.html
NFT Gallery
login.html
Authentication
apis.html
API Manager
security.html
Security Settings
services.html
Health Status
backup.html
Backup & Restore
logs.html
System Logs
cache.html
Cache Manager
styles.css
Theming (5 themes)
transactions.js
Transaction Analyzer
app.js
Client Logic
Transaction API
REST API
JSON
Backend (FastAPI + Uvicorn)
main.py
Application Entry
static
rate limit
request
session check
nft_image_database.py
Image Cache DB
config.py
Configuration
Security & Demo Middleware
rate_limit.py
Rate Limiting (slowapi)
demo_mode.py
Demo User Detection
size_limit.py
Request Size (10MB)
auth.py
Session Authentication
demo user
API keys
sessions
SSL config
SVG files
cert files
images
log entries
database.py
SQLite Layer
API Routers
/settings
API Key Management
/search
Search Endpoints
/system
System Health
/analytics
Analytics Endpoints
/auth
Login/Logout/Sessions
/security
SSL/Certificate Mgmt
/portfolio
Summary & History
/transactions
Multi-Chain Txn History
/prices
Crypto Prices
/nfts
NFT Collections
/custom-tokens
Token Tracking
/exchanges
42 Exchange APIs
+ ExchangeRegistry
/defi/protocols
Protocol Detection
+ P&L Analytics
/pnl
Cost Basis
P&L Reporting
/logs
System Logging
/defi
DeFi Positions
/images
Token Image Cache
/privacy
Privacy & ZK
Shielded Pool Stats
/wallets
Wallet CRUD
/backup
Export/Import
/nft-scheduler
Background NFT Updates
ExchangeRegistry
BSC/ARB/AVAX NFTs
shielded stats
Demo Services (Fake Data)
demo_wallet_service.py
91 Tokens, 26 Chains
demo_nft_service.py
76 NFTs, 7 Collections
demo_transaction_service.py
1,500 Transactions
demo_exchange_service.py
Exchange Balances
demo_defi_service.py
Staking Positions
Exchange Services (42 Exchanges via BaseExchangeService)
base_exchange.py
BaseExchangeService
+ 5 Auth Mixins
35 more
coinbase.py
Coinbase CDP
binance_service.py
Binance.com
binance_us_service.py
Binance.US
okx_service.py
OKX Exchange
bitget_service.py
Bitget Exchange
gate_service.py
Gate.io
kucoin_service.py
KuCoin
bybit, mexc, kraken, gemini
bitfinex, htx, bingx, phemex
woox, poloniex, cryptocom
bitstamp, bitmart + 22 more
(35 new exchange services)
DeFi Protocol Adapters (80+ Protocols)
registry.py
ProtocolRegistry
(auto-discovery)
cost_basis_engine.py
FIFO/LIFO/Average
P&L Analytics
base_adapter.py
ProtocolAdapter ABC
5 DetectionMethods
cardano/
Minswap, SundaeSwap V3
WingRiders, Splash, Djed
FluidTokens, Lenfi, MuesliSwap
Liqwid, Indigo, Strike
Surf, Iagon (13 adapters)
evm/
Aave v3, Compound v3
Uniswap v3 LP, Curve
Balancer, EigenLayer
Maker, Morpho, GMX
Pendle, Stargate, Aerodrome
Velodrome, Radiant, Benqi
SushiSwap, Yearn v3, Beefy
+ liquid staking tokens
55+ protocols
solana/
Marinade, Jito, Orca
Raydium, Drift, MarginFi
Kamino, Jupiter Perps
BlazeStake + 6 more
15 adapters
Support Services
ssl_service.py
Certificate Mgmt
snapshot.py
Portfolio Snapshots
logging_service.py
Centralized Logs
transaction_history.py
Multi-Chain Txns
pricing.py
Price Aggregation
defi.py
DeFi Protocols
taptools.py
Cardano Token Data
privacy_detector.py
Railgun Detection
(ETH/Polygon/BSC/ARB)
coinpaprika.py
Pricing Fallback
(free, 25k/month)
image_cache.py
Token Image Cache
etherscan.py
ETH Transaction History
moralis.py
Multi-Chain Fallback
logostream.py
Token Logo API
Services Layer
Blockchain Services (53 Chains)
monero.py
Monero (XMR)
(manual balance only)
vechain.py
VeChain (VET)
cosmos.py
Cosmos (ATOM)
near.py
NEAR (NEAR)
icp.py
ICP (ICP)
cosmos_chain.py
Cosmos IBC (config-driven)
Osmosis/Celestia/Injective
dYdX/Sei/Akash
substrate_service.py
Substrate Chains
Polkadot (DOT)
Kusama (KSM)
ton_service.py
TON (TON)
stellar_service.py
Stellar (XLM)
secret_network.py
Secret Network (SCRT)
kaspa_service.py
Kaspa (KAS)
ergo/iota/waves
mina/zilliqa/kaia
(public REST APIs)
cardano.py
Cardano (ADA)
bitcoin.py
Bitcoin (BTC)
ethereum.py
Ethereum (ETH)
polygon.py
Polygon (MATIC)
base.py
Base (ETH)
solana.py
Solana (SOL)
algorand.py
Algorand (ALGO)
evm_chain.py
Generic EVM Service
(BSC/Arbitrum/Avalanche
Optimism/zkSync/Linea
Scroll/Fantom/Cronos
Gnosis/Moonbeam/Kaia)
tron.py
Tron (TRX)
xrp.py
XRP (XRP)
hedera.py
Hedera (HBAR)
multiversx.py
MultiversX (EGLD)
sui.py
Sui (SUI)
aptos.py
Aptos (APT)
filecoin.py
Filecoin (FIL)
litecoin.py
Litecoin (LTC)
dogecoin.py
Dogecoin (DOGE)
zcash.py
Zcash (ZEC)
tezos.py
Tezos (XTZ)
stacks.py
Stacks (STX)
NFT Services
nft_image_service.py
Image Caching
nft.py
Cardano NFTs
ethereum_nft.py
Ethereum NFTs
solana_nft.py
Solana NFTs
algorand_nft.py
Algorand NFTs (ARC-3/69)
nft_scheduler.py
Background Updates
HTTPS
3rd
Chainlink Staking
2nd
HTTPS (fallback)
HTTPS (BSC/ARB/AVAX)
1st
5th
4th
HTTPS (35)
Cardano UTXOs
DeFi contract calls
Solana DeFi RPCs
Data Storage
static/demo-nfts/
Demo NFT SVGs
data/certs/
SSL Certificates
(server.crt, server.key)
nft_images.db
NFT Image Cache
(thumbnails, metadata)
portfolio.db
Main Database
(users, sessions, wallets,
balances, API keys, cache,
transactions, nfts,
security settings, logs)
Security & Monitoring
.git/hooks/pre-push
Git Integration
triggers
security_agent.py
Pre-push Hook
runs
security_audit.py
Automated Audit
External APIs
Exchange APIs (42 Platforms)
Coinbase CDP
Portfolio Sync
Binance.com
Global Exchange
Binance.US
US Exchange
OKX
Multi-Asset Exchange
Bitget
Derivatives Exchange
Gate.io
Comprehensive API
KuCoin
Global Exchange
35 New Exchanges
Bybit, MEXC, Kraken, Gemini
Bitfinex, HTX, Crypto.com
+ 28 more
Visual Assets
LogoKit
Blockchain Logos
Logostream
Token Logo API
Cardano Network
Blockfrost
Blockchain Data (Primary)
TapTools
NFT Prices & Token Data
Maestro
Blockchain Data (Fallback)
CExplorer
Staking Data
Koios
Fallback API
EVM Chains (12 via Alchemy + Public RPCs)
Arbiscan
Arbitrum Transactions
Snowscan
Avalanche Transactions
Etherscan
Ethereum Transactions
BscScan
BSC Transactions
Basescan
Base Transactions
Moralis
Multi-Chain Fallback
Polygonscan
Polygon Transactions
Alchemy
ETH/Polygon/Base/BSC/ARB/AVAX
+OPT/zkSync/Linea/Scroll
Algorand Network
Tatum
Algorand Data (Fallback)
Pera Wallet
Algorand Data (Primary)
Tron Network
TronGrid
TRX/TRC-20 Data (Free)
Other Blockchains
VeBlocks/Thor
VeChain Data (Free)
Cosmos LCD
PublicNode (Free)
NEAR RPC
+ NearBlocks (Free)
Rosetta API
ICP Data (Free)
Cosmos IBC LCD
Osmosis/Celestia/Injective
dYdX/Sei/Akash (Free)
Subscan
Polkadot/Kusama (opt. key)
TON Center
TON Data (opt. key)
Horizon API
Stellar Data (Free)
SecretSaturn LCD
Secret Network (Free)
Blockstream
Bitcoin Data
Mempool.space
Bitcoin Fallback
Helius
Solana RPC & NFTs
XRPL Cluster
XRP Ledger API
Mirror Node
Hedera API
MultiversX API
MultiversX Data
Sui RPC
Sui Data
Aptos API
Aptos Data
Glif RPC
Filecoin Data
BlockCypher
LTC/DOGE Data (Free)
Blockchair
Zcash Data (Free)
TzKT
Tezos Data (Free)
Hiro API
Stacks Data (Free)
Pricing (Fallback Chain)
3. Coinbase
Spot Prices (Free)
2. CoinMarketCap
Secondary Prices
1. CoinGecko
Primary Price Data
5. DefiLlama
Universal Fallback
4. CoinPaprika
Free Fallback (25k/mo)